

Term 3 Practice Activity: C1 Density Functions (No integration)

Contents

- 1 Checking Two Candidate Density Functions I (Trade Throughput)
- 2 Checking Two Candidate Density Functions II (Shipping Delay Index)
- 3 Checking Two Candidate Density Functions III (Customs Processing Measure)
- 4 Regional Export Volume (Uniform)
- 5 Port Clearance Delay (Linear, Endpoint at 0)
- 6 Commodity Quality Score (Linear, Both Endpoints Nonzero)
- 7 Freight Cost Band (Uniform)
- 8 Warehouse Processing Time (Linear, Endpoint at 0)
- 9 Supplier Reliability Index (Linear, Both Endpoints Nonzero)
- 10 Import Batch Size (Uniform)
- 11 Transaction Size (Linear, Endpoint at 0)
- 12 Tariff Compliance Score (Linear, Both Endpoints Nonzero)
- 13 Port Queue Waiting Time (Uniform)
- 14 Export Readiness Fraction (Linear, Endpoint at 0)
- 15 Regional Competitiveness Measure (Linear, Both Endpoints Nonzero)
- 16 Cross-Border Fee Spread (Uniform with Variable Domain)
- 17 Global Platform Usage Index (Linear with Variable Domain)
- 18 Factory Output Deviation (Linear with One Variable Endpoint)
- 19 Customs Inspection Time (Uniform with Two Variable Endpoints)
- 20 Market Participation Measure (Increasing Linear with Two Variable Endpoints)

21 Inflation Relief Level (Decreasing Linear with Two Variable Endpoints)

22 Cross-Border Delivery Deviation (Shifted Linear with Two Variable Endpoints)

Evaluated criteria

- C1.

1 Checking Two Candidate Density Functions I (Trade Throughput)

Problem. A continuous random variable describing a global-economics indicator is modeled on the interval shown in each case.

Distribution A

$$f(x) = 0.25 \quad \text{for } 0 \leq x \leq 4$$

Distribution B

$$f(x) = 0.40 \quad \text{for } 0 \leq x \leq 4$$

Questions/Tasks.

- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using the total area under each graph.
- **Extra analysis:** For the valid distribution, find $P(1 \leq X \leq 3)$.

C1

C1 calculations. For a valid PDF, $f(x) \geq 0$ on its support and total area is 1.

- Distribution A: support $[0, 4]$, constant height 0.25.

$$\text{Area} = 4(0.25) = 1,$$

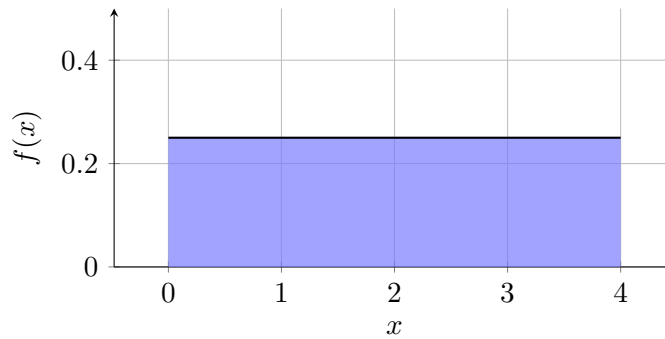
so A is valid.

- Distribution B: support $[0, 4]$, constant height 0.40.

$$\text{Area} = 4(0.40) = 1.6 \neq 1,$$

so B is not valid.

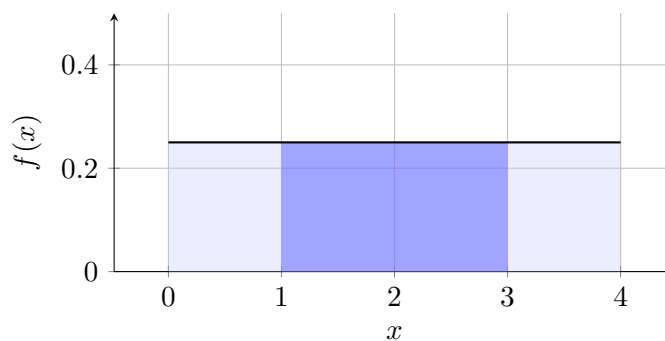
C2 interpretation.



The valid PDF is uniform on $[0, 4]$: it is constant, nonnegative, and gives equal probability to equal-width intervals.

Extra analysis. For the valid distribution A,

$$P(1 \leq X \leq 3) = (3 - 1)(0.25) = 0.5.$$



Final answer: A is valid, B is not valid, and $P(1 \leq X \leq 3) = 0.5$.

2 Checking Two Candidate Density Functions II (Shipping Delay Index)

Problem. A continuous random variable describing a global-economics indicator is modeled on the interval shown in each case.

Distribution A

$$f(x) = \frac{x}{8} \quad \text{for } 0 \leq x \leq 4$$

Distribution B

$$f(x) = \frac{x}{6} \quad \text{for } 0 \leq x \leq 4$$

Questions/Tasks.

- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using geometric reasoning about the area under each line.
- **Extra analysis:** For the valid distribution, find $P(2 \leq X \leq 4)$.

C1

C1 calculations. Check nonnegativity and total area on support $[0, 4]$.

- Distribution A: $f(x) = \frac{x}{8}$ forms a triangle with base 4 and height $\frac{4}{8} = \frac{1}{2}$.

$$\text{Area} = \frac{1}{2}(4) \left(\frac{1}{2} \right) = 1,$$

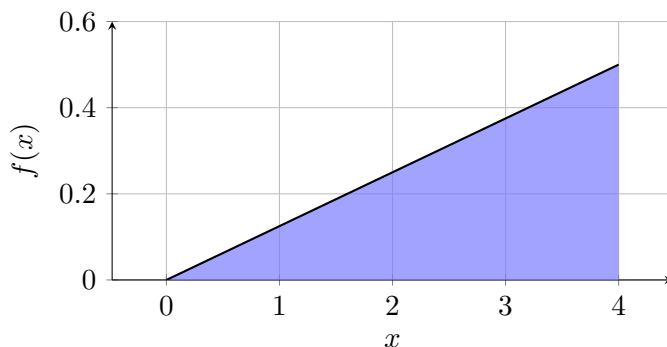
so A is valid.

- Distribution B: $f(x) = \frac{x}{6}$ has base 4 and height $\frac{4}{6} = \frac{2}{3}$.

$$\text{Area} = \frac{1}{2}(4) \left(\frac{2}{3} \right) = \frac{4}{3} \neq 1,$$

so B is not valid.

C2 interpretation.



The valid PDF increases linearly on $[0, 4]$, so larger x values have higher density than smaller ones.

Extra analysis. For A,

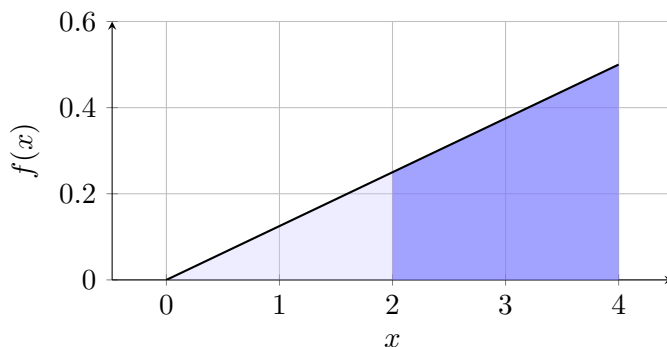
$$P(2 \leq X \leq 4) = 1 - P(0 \leq X \leq 2).$$

On $[0, 2]$, area is a triangle with base 2 and height $\frac{2}{8} = \frac{1}{4}$:

$$P(0 \leq X \leq 2) = \frac{1}{2}(2) \left(\frac{1}{4} \right) = \frac{1}{4}.$$

Hence

$$P(2 \leq X \leq 4) = 1 - \frac{1}{4} = \frac{3}{4}.$$



Final answer: A is valid, B is not valid, and $P(2 \leq X \leq 4) = \frac{3}{4}$.

3 Checking Two Candidate Density Functions III (Customs Processing Measure)

Problem. A continuous random variable describing a global-economics indicator is modeled on the interval shown in each case.

Distribution A

$$f(x) = \frac{6-x}{12} \quad \text{for } 1 \leq x \leq 5$$

Distribution B

$$f(x) = \frac{6-x}{10} \quad \text{for } 1 \leq x \leq 5$$

Questions/Tasks.

- **C1 Calculations:** Decide which distribution is a valid PDF and which one is not.
- **C1 Interpretation:** Justify your answer using geometric reasoning about the area under each line.
- **Extra analysis:** For the valid distribution, find $P(1 \leq X \leq 3)$.

C1

C1 calculations. On support $[1, 5]$, both densities are nonnegative. Check total area:

- Distribution A: $f(x) = \frac{6-x}{12}$ gives a trapezium with base 4, left height $f(1) = \frac{5}{12}$, and right height $f(5) = \frac{1}{12}$.

$$\text{Area} = \frac{\frac{5}{12} + \frac{1}{12}}{2} \cdot 4 = 1,$$

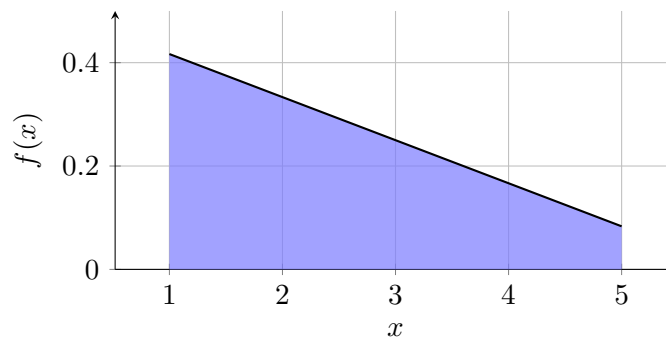
so A is valid.

- Distribution B: $f(x) = \frac{6-x}{10}$ gives base 4, left height $f(1) = \frac{5}{10} = \frac{1}{2}$, and right height $f(5) = \frac{1}{10}$.

$$\text{Area} = \frac{\frac{1}{2} + \frac{1}{10}}{2} \cdot 4 = \frac{6}{5} \neq 1,$$

so B is not valid.

C2 interpretation.



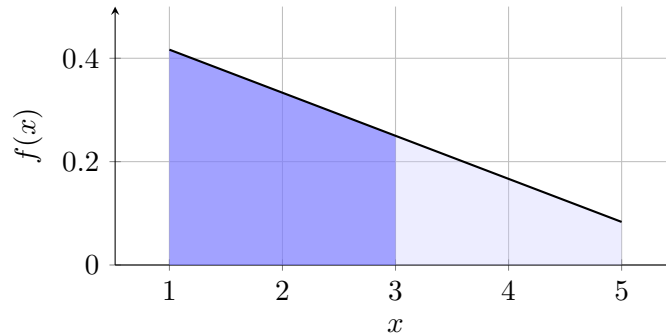
The valid PDF is decreasing on $[1, 5]$, so values near 1 are more likely than values near 5, but values near 5 still have positive density.

Extra analysis. For valid distribution A,

$$P(1 \leq X \leq 3) = \text{area of trapezium on } [1, 3].$$

With heights $f(1) = \frac{5}{12}$ and $f(3) = \frac{1}{4}$, width 2:

$$P(1 \leq X \leq 3) = \frac{\frac{5}{12} + \frac{1}{4}}{2} \cdot 2 = \frac{2}{3}.$$



Final answer: A is valid, B is not valid, and $P(1 \leq X \leq 3) = \frac{2}{3}$.

4 Regional Export Volume (Uniform)

Problem. Let X be the weekly export volume (in million units) with support $[0, 4]$ and density

$$f(x) = k \quad \text{for } 0 \leq x \leq 4.$$

Questions/Tasks.

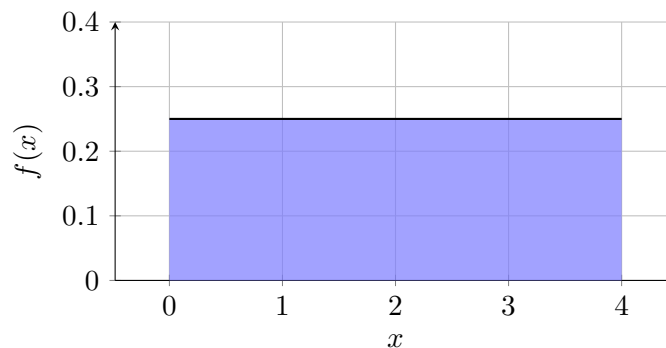
- **C1 Calculations:** Find k so $f(x)$ is a valid PDF.
- **C1 Interpretation:** Explain what this model says about equal-length intervals.
- **Extra analysis:** Find $P(1 \leq X \leq 3)$.

C1

C1 calculations. Since $f(x) = k$ on $[0, 4]$, total area is rectangle area:

$$4k = 1 \implies k = \frac{1}{4}.$$

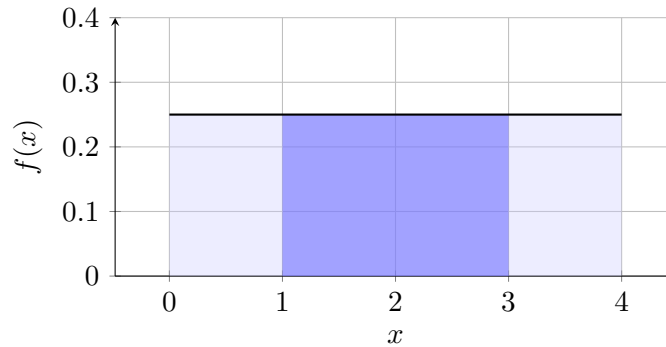
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For the requested probability,

$$P(1 \leq X \leq 3) = \text{width} \times \text{height} = (3 - 1) \left(\frac{1}{4} \right) = \frac{1}{2}.$$



Final answer: $k = \frac{1}{4}$ and $P(1 \leq X \leq 3) = \frac{1}{2}$.

5 Port Clearance Delay (Linear, Endpoint at 0)

Problem. Let T be a port-clearance delay (in hours) with support $[0, 6]$ and density

$$f(t) = kt \quad \text{for } 0 \leq t \leq 6.$$

Questions/Tasks.

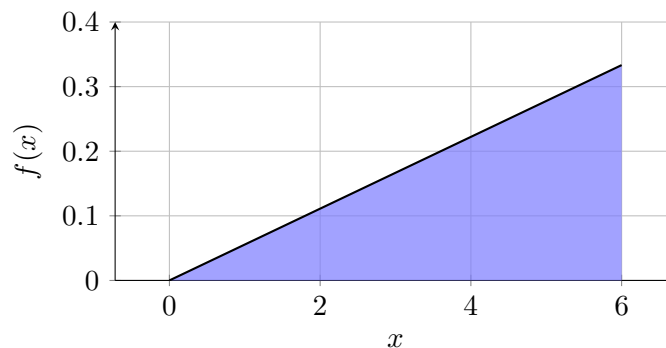
- **C1 Calculations:** Find k so $f(t)$ is a valid PDF.
- **C1 Interpretation:** Describe how probability changes as t increases.
- **Extra analysis:** Find $P(T \geq 4)$.

C1

C1 calculations. For $f(t) = kt$ on $[0, 6]$, area is a triangle with base 6 and height $6k$:

$$\frac{1}{2}(6)(6k) = 18k = 1 \implies k = \frac{1}{18}.$$

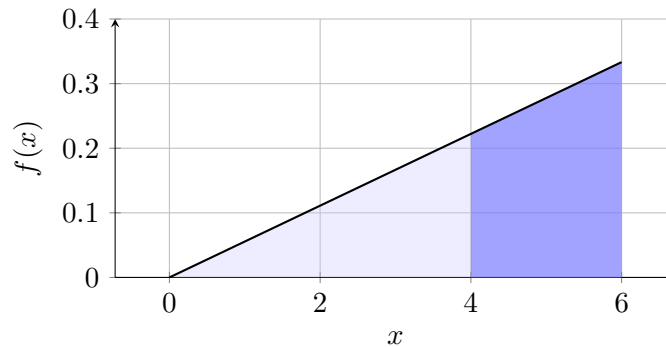
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For $P(T \geq 4)$, use area from 4 to 6 (a trapezium): heights are $4k$ and $6k$, width 2.

$$P(T \geq 4) = \frac{4k + 6k}{2} \cdot 2 = 10k = \frac{10}{18} = \frac{5}{9}.$$



Final answer: $k = \frac{1}{18}$ and $P(T \geq 4) = \frac{5}{9}$.

6 Commodity Quality Score (Linear, Both Endpoints Nonzero)

Problem. Let Q be a commodity quality score used in international procurement with support $[2, 8]$ and density

$$f(q) = k(q - 1) \quad \text{for } 2 \leq q \leq 8.$$

Questions/Tasks.

- **C1 Calculations:** Find k so $f(q)$ is a valid PDF.
- **C1 Interpretation:** Compare density near $q = 3$ and $q = 7$.
- **Extra analysis:** Find $P(4 \leq Q \leq 6)$.

C1

C1 calculations. On $[2, 8]$, $f(q) = k(q - 1)$ is linear and positive. Total area is a trapezium with width 6, heights

$$f(2) = k, \quad f(8) = 7k.$$

So

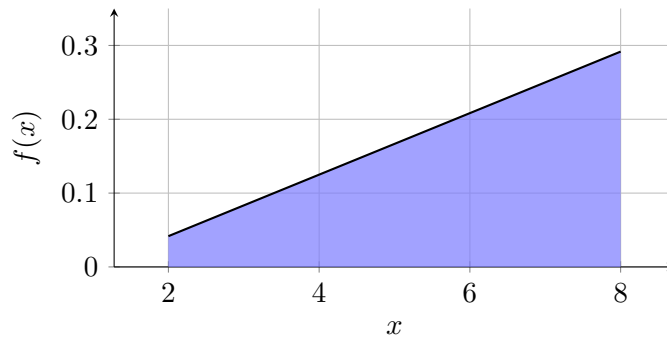
$$\frac{k + 7k}{2} \cdot 6 = 24k = 1 \implies k = \frac{1}{24}.$$

Compare densities:

$$f(3) = 2k, \quad f(7) = 6k,$$

so density near $q = 7$ is three times density near $q = 3$.

C2 interpretation.



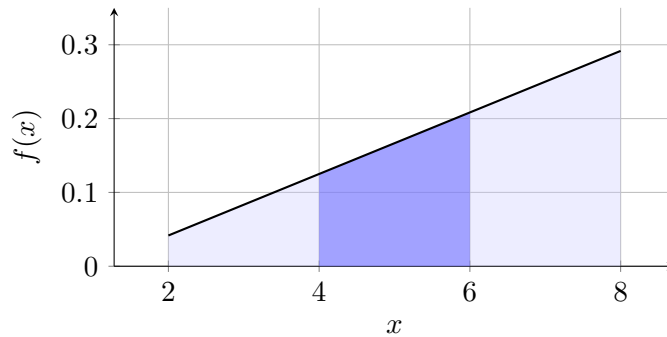
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For

$$P(4 \leq Q \leq 6) = \text{trapezium area on } [4, 6],$$

with heights $f(4) = 3k$, $f(6) = 5k$, width 2:

$$P(4 \leq Q \leq 6) = \frac{3k + 5k}{2} \cdot 2 = 8k = \frac{1}{3}.$$



Final answer: $k = \frac{1}{24}$ and $P(4 \leq Q \leq 6) = \frac{1}{3}$.

7 Freight Cost Band (Uniform)

Problem. Let Y be the freight cost deviation index across regions with support $[-2, 3]$ and density

$$f(y) = k \quad \text{for } -2 \leq y \leq 3.$$

Questions/Tasks.

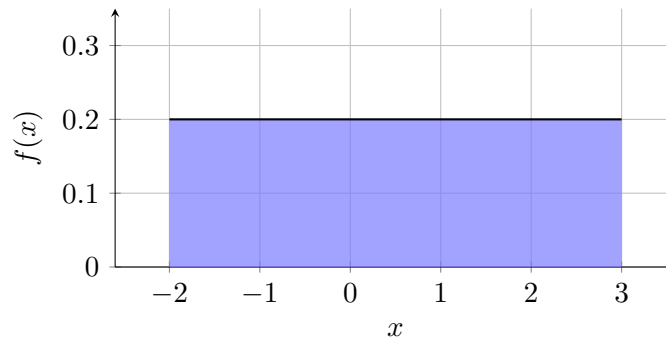
- **C1 Calculations:** Find k so $f(y)$ is a valid PDF.
- **C1 Interpretation:** Explain why interval length controls probability here.
- **Extra analysis:** Find $P(-1 \leq Y \leq 2)$.

C1

C1 calculations. Uniform on $[-2, 3]$ gives length 5, so

$$5k = 1 \implies k = \frac{1}{5}.$$

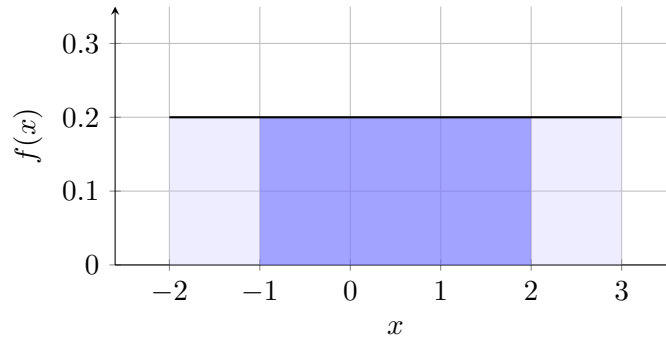
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Since density is constant, probability depends only on interval length:

$$P(-1 \leq Y \leq 2) = (2 - (-1)) \cdot \frac{1}{5} = \frac{3}{5}.$$



Final answer: $k = \frac{1}{5}$ and $P(-1 \leq Y \leq 2) = \frac{3}{5}$.

8 Warehouse Processing Time (Linear, Endpoint at 0)

Problem. Let A be warehouse processing time (in hours) with support $[0, 5]$ and density

$$f(a) = k(5 - a) \quad \text{for } 0 \leq a \leq 5.$$

Questions/Tasks.

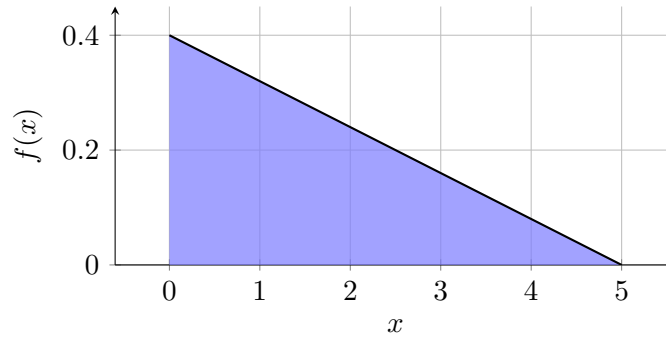
- **C1 Calculations:** Find k so $f(a)$ is a valid PDF.
- **C1 Interpretation:** Explain what the decreasing line says about short vs long times.
- **Extra analysis:** Find $P(1 \leq A \leq 4)$.

C1

C1 calculations. $f(a) = k(5 - a)$ on $[0, 5]$ gives a decreasing line from height $5k$ to 0. Total area (triangle):

$$\frac{1}{2}(5)(5k) = \frac{25k}{2} = 1 \implies k = \frac{2}{25}.$$

C2 interpretation.



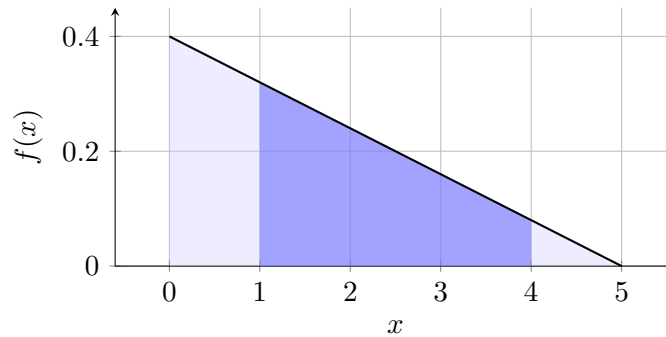
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For $1 \leq A \leq 4$, use trapezium area with heights

$$f(1) = 4k, \quad f(4) = k,$$

and width 3:

$$P(1 \leq A \leq 4) = \frac{4k + k}{2} \cdot 3 = \frac{15k}{2} = \frac{3}{5}.$$



Final answer: $k = \frac{2}{25}$ and $P(1 \leq A \leq 4) = \frac{3}{5}$.

9 Supplier Reliability Index (Linear, Both Endpoints Nonzero)

Problem. Let R be a supplier reliability index in cross-border contracts with support $[1, 7]$ and density

$$f(r) = k(r + 1) \quad \text{for } 1 \leq r \leq 7.$$

Questions/Tasks.

- **C1 Calculations:** Find k so $f(r)$ is a valid PDF.
- **C1 Interpretation:** Explain why right-side intervals have more probability than equal-width left-side intervals.
- **Extra analysis:** Find $P(R \leq 4)$.

C1

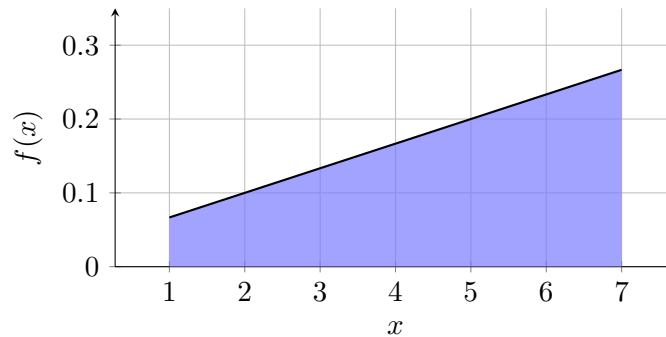
C1 calculations. On $[1, 7]$, $f(r) = k(r + 1)$ has

$$f(1) = 2k, \quad f(7) = 8k.$$

Total area (trapezium, width 6):

$$\frac{2k + 8k}{2} \cdot 6 = 30k = 1 \implies k = \frac{1}{30}.$$

C2 interpretation.



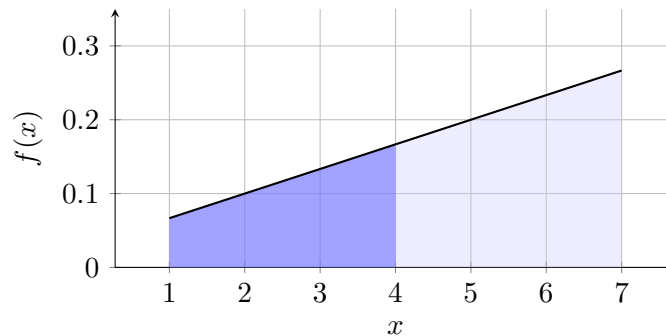
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For

$$P(R \leq 4) = P(1 \leq R \leq 4) = \frac{f(1) + f(4)}{2} \cdot (4 - 1),$$

with $f(4) = 5k$:

$$P(R \leq 4) = \frac{2k + 5k}{2} \cdot 3 = \frac{21k}{2} = \frac{7}{20}.$$



Final answer: $k = \frac{1}{30}$ and $P(R \leq 4) = \frac{7}{20}$.

10 Import Batch Size (Uniform)

Problem. Let M be import batch size (in tons) with support $[3, b]$ and density

$$f(m) = \frac{1}{6} \quad \text{for } 3 \leq m \leq b.$$

Questions/Tasks.

- **C1 Calculations:** Find b so $f(m)$ is a valid PDF.
- **C1 Interpretation:** State two different intervals with equal probability and justify.
- **Extra analysis:** Find $P(4.5 \leq M \leq 7.5)$.

C1

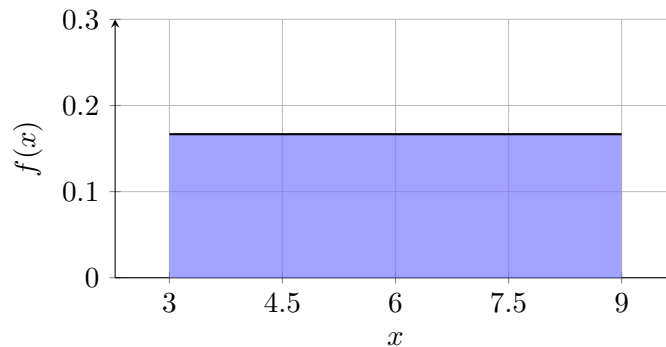
C1 calculations. The height is fixed at $\frac{1}{6}$, so the total area must be

$$(b - 3) \cdot \frac{1}{6} = 1.$$

Hence

$$b - 3 = 6 \implies b = 9.$$

C2 interpretation.

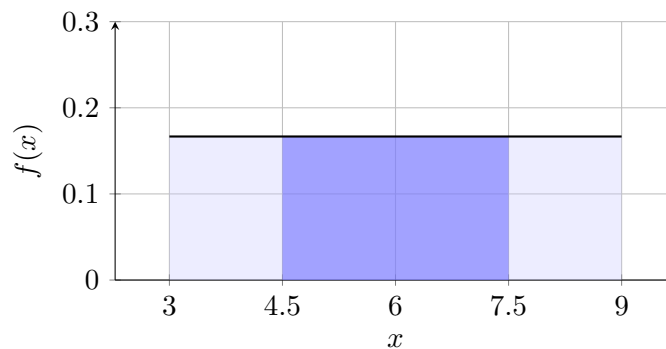


The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Two equal-length intervals have equal probability, for example:

$$P(3 \leq M \leq 5) = P(7 \leq M \leq 9) = 2 \cdot \frac{1}{6} = \frac{1}{3}.$$

$$P(4.5 \leq M \leq 7.5) = (7.5 - 4.5) \cdot \frac{1}{6} = \frac{3}{6} = \frac{1}{2}.$$



Final answer: $b = 9$, equal-length intervals such as $[3, 5]$ and $[7, 9]$ have equal probability, and $P(4.5 \leq M \leq 7.5) = \frac{1}{2}$.

11 Transaction Size (Linear, Endpoint at 0)

Problem. Let B be a cross-border transaction-size index with support $[0, c]$ and density

$$f(b) = \frac{1}{16} + \frac{b}{8c} \quad \text{for } 0 \leq b \leq c.$$

Questions/Tasks.

- **C1 Calculations:** Find c so $f(b)$ is a valid PDF.
- **C1 Interpretation:** Compare likelihood of values in $[0, 2]$ and $[6, 8]$.
- **Extra analysis:** Find $P(2 \leq B \leq 6)$.

C1

C1 calculations. The endpoint values are fixed:

$$f(0) = \frac{1}{16}, \quad f(c) = \frac{1}{16} + \frac{c}{8c} = \frac{3}{16}.$$

Total area must be 1. Using the trapezium formula,

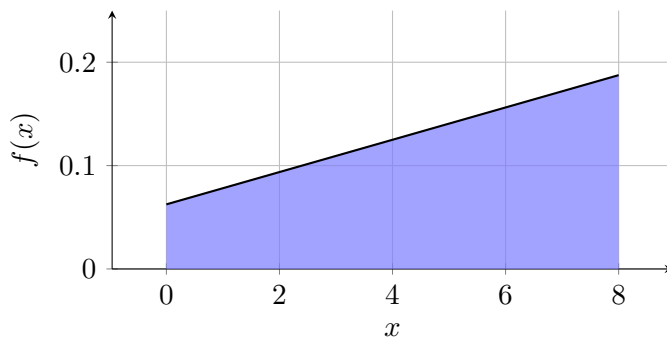
$$\frac{\frac{1}{16} + \frac{3}{16}}{2} \cdot c = 1.$$

So

$$\frac{4/16}{2} \cdot c = 1$$

$$\frac{1}{8}c = 1 \implies c = 8.$$

C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Using $c = 8$, the density is

$$f(b) = \frac{1}{16} + \frac{b}{64} \quad \text{for } 0 \leq b \leq 8.$$

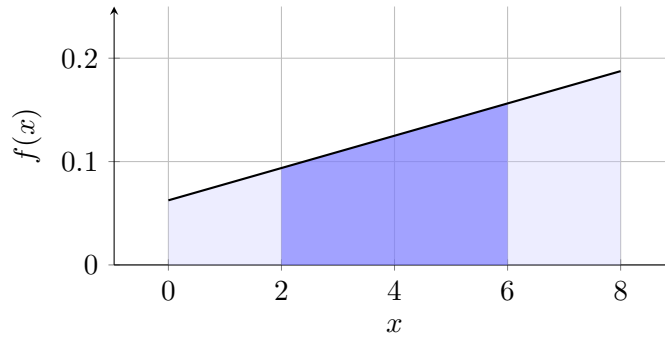
For the comparison:

$$P(0 \leq B \leq 2) = \frac{\frac{1}{16} + \frac{3}{32}}{2} \cdot 2 = \frac{5}{32}, \quad P(6 \leq B \leq 8) = \frac{\frac{5}{32} + \frac{3}{16}}{2} \cdot 2 = \frac{11}{32},$$

so $[6, 8]$ is more likely.

For the requested probability,

$$P(2 \leq B \leq 6) = \frac{f(2) + f(6)}{2} \cdot 4 = \frac{\frac{3}{32} + \frac{5}{32}}{2} \cdot 4 = \frac{1}{2}.$$



Final answer: $c = 8$ and $P(2 \leq B \leq 6) = \frac{1}{2}$.

12 Tariff Compliance Score (Linear, Both Endpoints Nonzero)

Problem. Let D be a tariff compliance score with support $[-3, c]$ and density

$$f(d) = \frac{1}{40} + \frac{d+3}{5(c+3)} \quad \text{for } -3 \leq d \leq c.$$

Questions/Tasks.

- **C1 Calculations:** Find c so $f(d)$ is a valid PDF.
- **C1 Interpretation:** Explain why the model is nonnegative on its support.
- **Extra analysis:** Find $P(-1 \leq D \leq 3)$.

C1

C1 calculations. The endpoint values are fixed:

$$f(-3) = \frac{1}{40}, \quad f(c) = \frac{1}{40} + \frac{c+3}{5(c+3)} = \frac{1}{40} + \frac{1}{5} = \frac{9}{40}.$$

Since the graph is a line joining two positive endpoint values, the density is nonnegative on its support.

The total area must be 1. Using the trapezium formula,

$$\frac{\frac{1}{40} + \frac{9}{40}}{2} \cdot (c+3) = 1.$$

So

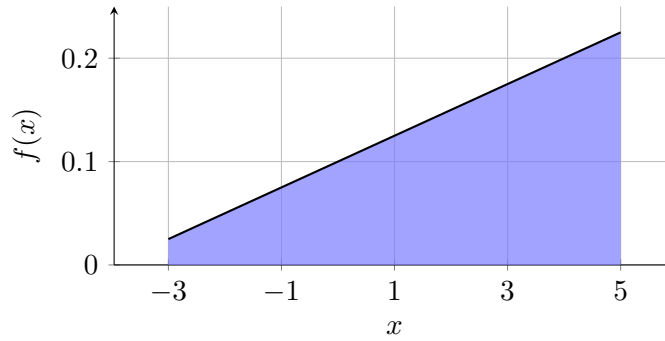
$$\frac{10/40}{2} \cdot (c+3) = 1$$

$$\frac{1}{8}(c+3) = 1 \implies c+3 = 8 \implies c = 5.$$

Now substitute this value into the density:

$$f(d) = \frac{1}{40} + \frac{d+3}{40} = \frac{d+4}{40} \quad \text{for } -3 \leq d \leq 5.$$

C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For the requested probability,

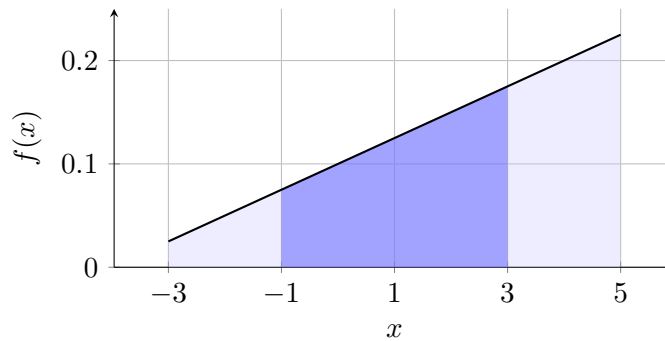
$$P(-1 \leq D \leq 3) = \frac{f(-1) + f(3)}{2} \cdot 4.$$

With

$$f(-1) = \frac{3}{40}, \quad f(3) = \frac{7}{40},$$

we get

$$P(-1 \leq D \leq 3) = \frac{\frac{3}{40} + \frac{7}{40}}{2} \cdot 4 = \frac{1}{2}.$$



Final answer: $c = 5$, the density is nonnegative since both endpoint heights are positive and the graph is linear on the support, and $P(-1 \leq D \leq 3) = \frac{1}{2}$.

13 Port Queue Waiting Time (Uniform)

Problem. Let W be port queue waiting time (in minutes) with support $[2, c]$. The density is a line on this support, with endpoint values

$$f(2) = \frac{1}{8} \quad \text{and} \quad f(c) = \frac{1}{8}.$$

Questions/Tasks.

- **C1 Calculations:** Find c so the model is a valid PDF.
- **C1 Interpretation:** Explain how to compare probabilities of two equal-width waiting-time intervals.
- **Extra analysis:** Find $P(W > 7)$.

C1

C1 calculations. Since the density is a line and has the same value at both endpoints, it is constant on the support. So the graph is a rectangle of height $\frac{1}{8}$ and width $c - 2$.

For a valid PDF, the total area must be 1:

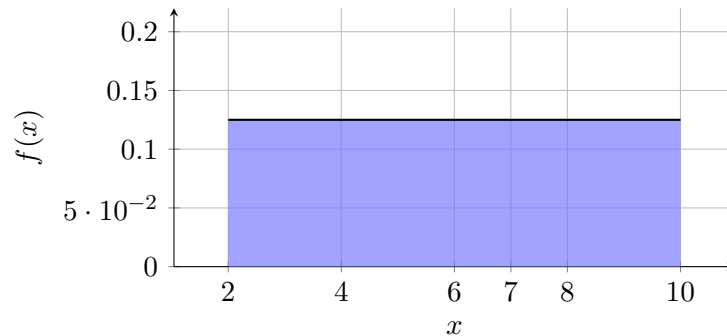
$$(c - 2) \cdot \frac{1}{8} = 1.$$

Thus

$$c - 2 = 8 \implies c = 10.$$

For a constant density, equal-width intervals have equal probability because probability is width $\times \frac{1}{8}$.

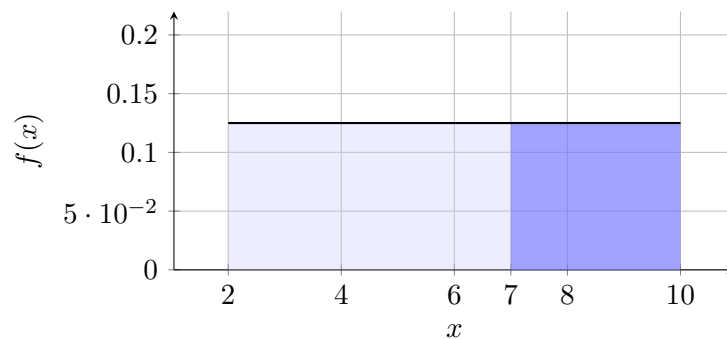
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis.

$$P(W > 7) = P(7 < W \leq 10) = (10 - 7) \cdot \frac{1}{8} = \frac{3}{8}.$$



Final answer: $c = 10$ and $P(W > 7) = \frac{3}{8}$.

14 Export Readiness Fraction (Linear, Endpoint at 0)

Problem. Let C be an export-readiness fraction of firms with support $[0, b]$. The density is a line on this support, with endpoint values

$$f(0) = 1 \quad \text{and} \quad f(b) = 0.$$

Questions/Tasks.

- **C1 Calculations:** Find b so the model is a valid PDF.
- **C1 Interpretation:** Explain the practical meaning of higher density near $c = 0$ than near the right endpoint.
- **Extra analysis:** Find $P(\frac{1}{2} \leq C \leq \frac{3}{2})$.

C1

C1 calculations. Since the density is a line with endpoint values $f(0) = 1$ and $f(b) = 0$, the graph is a decreasing triangle with base b and height 1.

For a valid PDF, the total area must be 1:

$$\frac{1}{2}(b)(1) = 1.$$

Thus

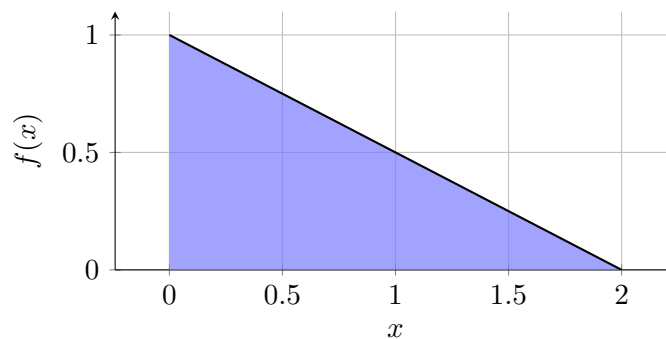
$$b = 2.$$

So the density is the line joining $(0, 1)$ and $(2, 0)$, that is,

$$f(c) = 1 - \frac{c}{2} \quad \text{for } 0 \leq c \leq 2.$$

Higher density near $c = 0$ means lower completion fractions occur more often than values near full completion.

C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For

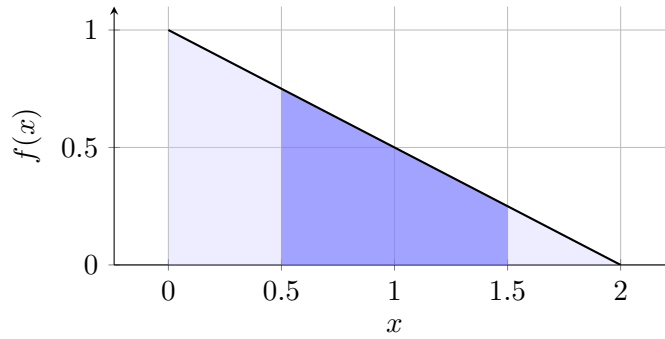
$$\frac{1}{2} \leq C \leq \frac{3}{2},$$

use trapezium area. The endpoint heights are

$$f\left(\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}, \quad f\left(\frac{3}{2}\right) = 1 - \frac{3}{4} = \frac{1}{4},$$

with width 1. So

$$P\left(\frac{1}{2} \leq C \leq \frac{3}{2}\right) = \frac{\frac{3}{4} + \frac{1}{4}}{2} \cdot 1 = \frac{1}{2}.$$



Final answer: $b = 2$ and $P(\frac{1}{2} \leq C \leq \frac{3}{2}) = \frac{1}{2}$.

15 Regional Competitiveness Measure (Linear, Both Endpoints Nonzero)

Problem. Let P be a regional competitiveness measure with support $[1, c]$. The density is a line on this support, with endpoint values

$$f(1) = \frac{9}{40} \quad \text{and} \quad f(c) = \frac{1}{40}.$$

Questions/Tasks.

- **C1 Calculations:** Find c so the model is a valid PDF.
- **C1 Interpretation:** Compare $P(1 \leq P \leq 3)$ and $P(7 \leq P \leq 9)$ without integration, using geometric reasoning.
- **Extra analysis:** Find $P(3 \leq P \leq 8)$.

C1

C1 calculations. Since the density is a line with endpoint values

$$f(1) = \frac{9}{40}, \quad f(c) = \frac{1}{40},$$

the graph is a decreasing trapezium on $[1, c]$.

For a valid PDF, the total area must be 1:

$$\frac{\frac{9}{40} + \frac{1}{40}}{2} \cdot (c - 1) = 1.$$

Thus

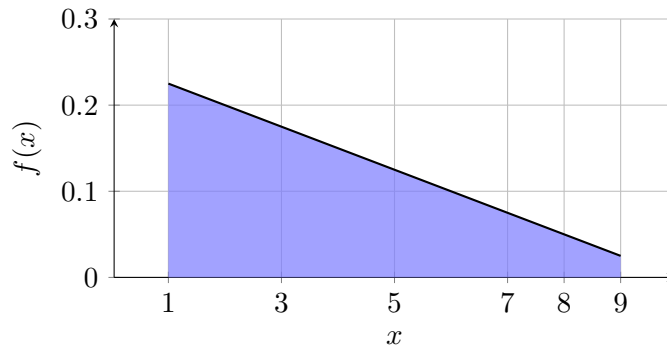
$$\frac{10/40}{2} \cdot (c - 1) = 1$$

$$\frac{1}{8}(c - 1) = 1 \implies c - 1 = 8 \implies c = 9.$$

So the density is the line joining $(1, \frac{9}{40})$ and $(9, \frac{1}{40})$, that is,

$$f(p) = \frac{10 - p}{40} \quad \text{for } 1 \leq p \leq 9.$$

C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Compare intervals geometrically:

$$P(1 \leq P \leq 3) = \frac{f(1) + f(3)}{2} \cdot 2$$

with

$$f(1) = \frac{9}{40}, \quad f(3) = \frac{7}{40}.$$

So

$$P(1 \leq P \leq 3) = \frac{\frac{9}{40} + \frac{7}{40}}{2} \cdot 2 = \frac{16}{40} = \frac{2}{5}.$$

Also,

$$P(7 \leq P \leq 9) = \frac{f(7) + f(9)}{2} \cdot 2$$

with

$$f(7) = \frac{3}{40}, \quad f(9) = \frac{1}{40}.$$

So

$$P(7 \leq P \leq 9) = \frac{\frac{3}{40} + \frac{1}{40}}{2} \cdot 2 = \frac{4}{40} = \frac{1}{10}.$$

Therefore,

$$P(1 \leq P \leq 3) > P(7 \leq P \leq 9).$$

For $3 \leq P \leq 8$:

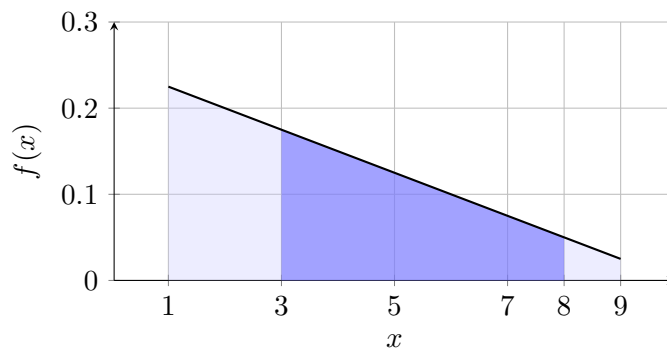
$$P(3 \leq P \leq 8) = \frac{f(3) + f(8)}{2} \cdot 5$$

with

$$f(3) = \frac{7}{40}, \quad f(8) = \frac{2}{40}.$$

Thus

$$P(3 \leq P \leq 8) = \frac{\frac{7}{40} + \frac{2}{40}}{2} \cdot 5 = \frac{9}{16}.$$



Final answer: $c = 9$, $P(1 \leq P \leq 3) = \frac{2}{5} > P(7 \leq P \leq 9) = \frac{1}{10}$, and $P(3 \leq P \leq 8) = \frac{9}{16}$.

16 Cross-Border Fee Spread (Uniform with Variable Domain)

Problem. Let S be a cross-border fee spread index with support $[0, a]$, where $a > 0$, and density

$$f(s) = k \quad \text{for } 0 \leq s \leq a.$$

Questions/Tasks.

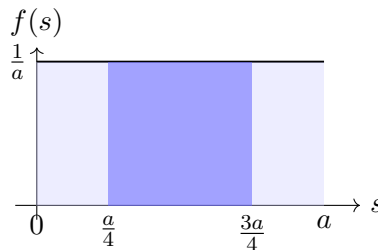
- **C1 Calculations:** Find k in terms of a so $f(s)$ is a valid PDF.
- **C1 Interpretation:** Explain what this model says about equal-length intervals inside $[0, a]$.
- **Extra analysis:** Find $P(\frac{a}{4} \leq S \leq \frac{3a}{4})$.

C1

C1 calculations. Uniform on $[0, a]$ with $a > 0$ gives

$$ak = 1 \implies k = \frac{1}{a}.$$

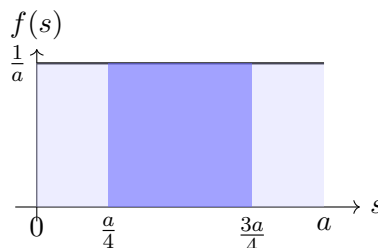
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For $\frac{a}{4} \leq S \leq \frac{3a}{4}$:

$$P\left(\frac{a}{4} \leq S \leq \frac{3a}{4}\right) = \left(\frac{3a}{4} - \frac{a}{4}\right) \frac{1}{a} = \frac{a/2}{a} = \frac{1}{2}.$$



Final answer: $k = \frac{1}{a}$ and $P(\frac{a}{4} \leq S \leq \frac{3a}{4}) = \frac{1}{2}$.

17 Global Platform Usage Index (Linear with Variable Domain)

Problem. Let F be a global platform-usage intensity index with support $[0, b]$, where $b > 0$, and density

$$f(f) = kf \quad \text{for } 0 \leq f \leq b.$$

Questions/Tasks.

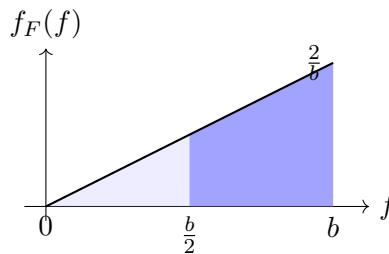
- **C1 Calculations:** Find k in terms of b so $f(f)$ is a valid PDF.
- **C1 Interpretation:** Explain why larger values of f are more likely than smaller values in equal-width intervals.
- **Extra analysis:** Find $P(\frac{b}{2} \leq F \leq b)$.

C1

C1 calculations. For $f(f) = kf$ on $[0, b]$ ($b > 0$), area is a triangle with base b and height kb :

$$\frac{1}{2}(b)(kb) = \frac{kb^2}{2} = 1 \implies k = \frac{2}{b^2}.$$

C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For the requested probability,

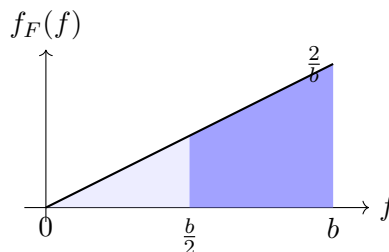
$$P\left(\frac{b}{2} \leq F \leq b\right) = 1 - P\left(0 \leq F \leq \frac{b}{2}\right).$$

Left part is a triangle with base $\frac{b}{2}$ and height $k\frac{b}{2}$:

$$P\left(0 \leq F \leq \frac{b}{2}\right) = \frac{1}{2} \left(\frac{b}{2}\right) \left(k\frac{b}{2}\right) = \frac{kb^2}{8} = \frac{1}{4}.$$

Hence

$$P\left(\frac{b}{2} \leq F \leq b\right) = 1 - \frac{1}{4} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{b^2}$ and $P(\frac{b}{2} \leq F \leq b) = \frac{3}{4}$.

18 Factory Output Deviation (Linear with One Variable Endpoint)

Problem. Let Z be a factory output deviation measure with support $[2, c]$, where $c > 2$, and density

$$f(z) = k(z - 2) \quad \text{for } 2 \leq z \leq c.$$

Questions/Tasks.

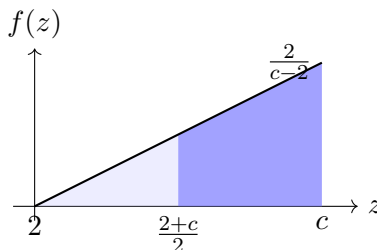
- **C1 Calculations:** Find k in terms of c so $f(z)$ is a valid PDF.
- **C1 Interpretation:** Explain why values near c are more likely than values near 2.
- **Extra analysis:** Find $P\left(\frac{2+c}{2} \leq Z \leq c\right)$ using geometric reasoning.

C1

C1 calculations. For $f(z) = k(z - 2)$ on $[2, c]$ ($c > 2$), area is a triangle with base $(c - 2)$ and height $k(c - 2)$:

$$\frac{1}{2}(c - 2)k(c - 2) = \frac{k(c - 2)^2}{2} = 1 \implies k = \frac{2}{(c - 2)^2}.$$

C2 interpretation.



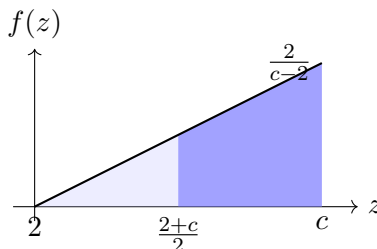
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Let midpoint be $\frac{2+c}{2}$. On this interval,

$$f\left(\frac{2+c}{2}\right) = k\frac{c-2}{2}, \quad f(c) = k(c-2), \quad \text{width} = \frac{c-2}{2}.$$

So

$$P\left(\frac{2+c}{2} \leq Z \leq c\right) = \frac{k\frac{c-2}{2} + k(c-2)}{2} \cdot \frac{c-2}{2} = \frac{3k(c-2)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(c-2)^2}$ and $P\left(\frac{2+c}{2} \leq Z \leq c\right) = \frac{3}{4}$.

19 Customs Inspection Time (Uniform with Two Variable Endpoints)

Problem. Let I be a customs inspection time with support $[a, b]$, where $a < b$, and density

$$f(i) = k \quad \text{for } a \leq i \leq b.$$

Questions/Tasks.

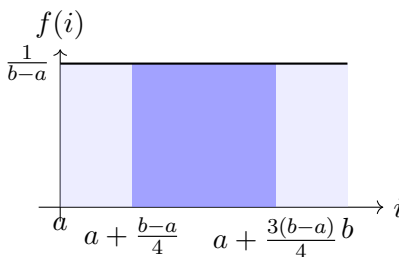
- **C1 Calculations:** Find k in terms of a and b so $f(i)$ is a valid PDF.
- **C1 Interpretation:** Explain why any two equal-length intervals inside $[a, b]$ have the same probability.
- **Extra analysis:** Find $P\left(a + \frac{b-a}{4} \leq I \leq a + \frac{3(b-a)}{4}\right)$.

C1

C1 calculations. Uniform on $[a, b]$ with $a < b$ gives support length $(b - a)$, so

$$k(b - a) = 1 \implies k = \frac{1}{b - a}.$$

C2 interpretation.



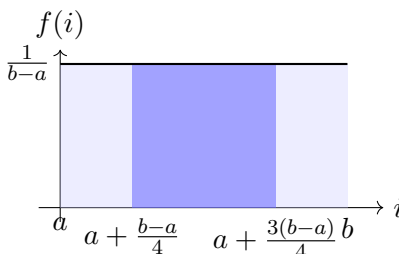
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For $a + \frac{b-a}{4} \leq I \leq a + \frac{3(b-a)}{4}$, interval length is

$$\left(a + \frac{3(b-a)}{4}\right) - \left(a + \frac{b-a}{4}\right) = \frac{b-a}{2}.$$

Hence

$$P = \frac{b-a}{2} \cdot \frac{1}{b-a} = \frac{1}{2}.$$



Final answer: $k = \frac{1}{b-a}$ and $P\left(a + \frac{b-a}{4} \leq I \leq a + \frac{3(b-a)}{4}\right) = \frac{1}{2}.$

20 Market Participation Measure (Increasing Linear with Two Variable End-points)

Problem. Let L be a market participation measure across trading blocs with support $[m, n]$, where $m < n$, and density

$$f(l) = k(l - m) \quad \text{for } m \leq l \leq n.$$

Questions/Tasks.

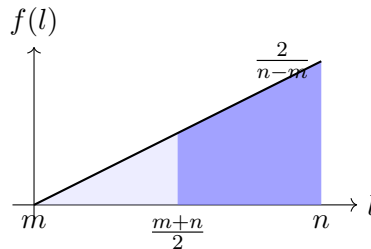
- **C1 Calculations:** Find k in terms of m and n so $f(l)$ is a valid PDF.
- **C1 Interpretation:** Explain why right-side intervals of the same width have more probability than left-side intervals.
- **Extra analysis:** Find $P\left(\frac{m+n}{2} \leq L \leq n\right)$ using geometric reasoning.

C1

C1 calculations. For $f(l) = k(l - m)$ on $[m, n]$ ($m < n$), this is a triangle from height 0 to height $k(n - m)$.

$$\frac{1}{2}(n - m)k(n - m) = \frac{k(n - m)^2}{2} = 1 \implies k = \frac{2}{(n - m)^2}.$$

C2 interpretation.



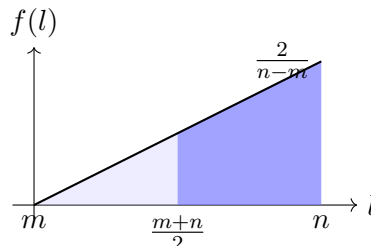
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For midpoint interval $\frac{m+n}{2} \leq L \leq n$:

$$f\left(\frac{m+n}{2}\right) = k\frac{n-m}{2}, \quad f(n) = k(n-m), \quad \text{width} = \frac{n-m}{2}.$$

So

$$P\left(\frac{m+n}{2} \leq L \leq n\right) = \frac{k\frac{n-m}{2} + k(n-m)}{2} \cdot \frac{n-m}{2} = \frac{3k(n-m)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(n-m)^2}$ and $P\left(\frac{m+n}{2} \leq L \leq n\right) = \frac{3}{4}$.

21 Inflation Relief Level (Decreasing Linear with Two Variable Endpoints)

Problem. Let C be an inflation-relief level index with support $[r, s]$, where $r < s$, and density

$$f(c) = k(s - c) \quad \text{for } r \leq c \leq s.$$

Questions/Tasks.

- **C1 Calculations:** Find k in terms of r and s so $f(c)$ is a valid PDF.
- **C1 Interpretation:** Explain why values near r are more likely than values near s .
- **Extra analysis:** Find $P(r \leq C \leq \frac{r+s}{2})$ using geometric reasoning.

C1

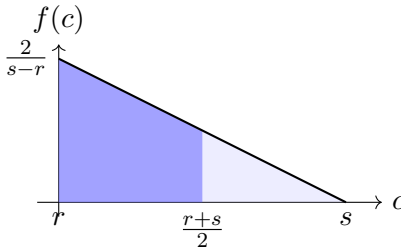
C1 calculations. For $f(c) = k(s - c)$ on $[r, s]$ ($r < s$), heights are

$$f(r) = k(s - r), \quad f(s) = 0,$$

so total area (triangle) is

$$\frac{1}{2}(s - r)k(s - r) = \frac{k(s - r)^2}{2} = 1 \implies k = \frac{2}{(s - r)^2}.$$

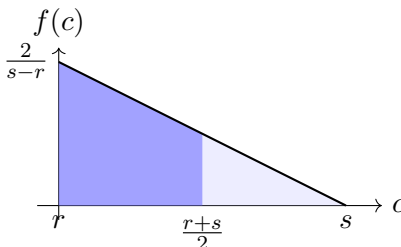
C2 interpretation.



The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. For $r \leq C \leq \frac{r+s}{2}$, width is $\frac{s-r}{2}$ and heights are $k(s - r)$ and $k\frac{s-r}{2}$.

$$P\left(r \leq C \leq \frac{r+s}{2}\right) = \frac{k(s - r) + k\frac{s-r}{2}}{2} \cdot \frac{s - r}{2} = \frac{3k(s - r)^2}{8} = \frac{3}{4}.$$



Final answer: $k = \frac{2}{(s-r)^2}$ and $P(r \leq C \leq \frac{r+s}{2}) = \frac{3}{4}$.

22 Cross-Border Delivery Deviation (Shifted Linear with Two Variable End-points)

Problem. Let D be a cross-border delivery deviation index with support $[p, q]$, where $p < q$, and density

$$f(d) = k(d - p + 1) \quad \text{for } p \leq d \leq q.$$

Questions/Tasks.

- **C1 Calculations:** Find k in terms of p and q so $f(d)$ is a valid PDF.
- **C1 Interpretation:** Explain why the density stays positive throughout the support.
- **Extra analysis:** Find $P(p + \frac{q-p}{3} \leq D \leq q)$.

C1

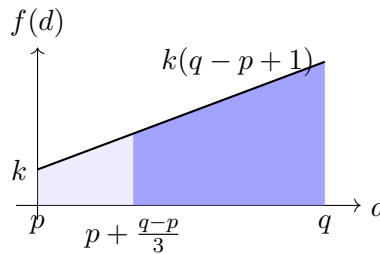
C1 calculations. For $f(d) = k(d - p + 1)$ on $[p, q]$ ($p < q$):

$$d - p + 1 \in [1, q - p + 1],$$

so density is positive on the whole support. Let $t = q - p > 0$. Then heights are k and $k(t + 1)$, width t , so

$$\frac{k + k(t + 1)}{2} \cdot t = \frac{k(t + 2)t}{2} = 1 \implies k = \frac{2}{(q - p)(q - p + 2)}.$$

C2 interpretation.



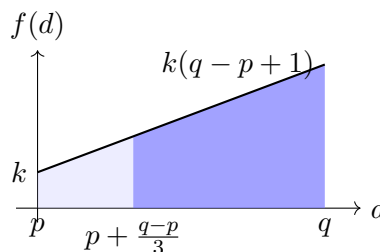
The PDF graph shows the support and overall shape of this global-economics density model.

Extra analysis. Now compute $P(p + \frac{q-p}{3} \leq D \leq q)$. Using $y = d - p$ ($0 \leq y \leq t$), this is $y \in [\frac{t}{3}, t]$ with density $k(y + 1)$:

$$P = \int_{t/3}^t k(y + 1) dy = \frac{k}{2} [(y + 1)^2]_{t/3}^t = \frac{k}{2} \left((t + 1)^2 - \left(\frac{t}{3} + 1 \right)^2 \right).$$

Substitute $k = \frac{2}{t(t+2)}$ and simplify:

$$P = \frac{4(2t + 3)}{9(t + 2)} = \frac{4(2(q - p) + 3)}{9(q - p + 2)}.$$



Final answer: $k = \frac{2}{(q-p)(q-p+2)}$ and

$$P\left(p + \frac{q-p}{3} \leq D \leq q\right) = \frac{4(2(q-p)+3)}{9(q-p+2)}.$$